A3-PS028-TC224

Happy Eyeballs IPv4/IPv6 Fallback Measurement

A controlled, evidence-driven study of dual-stack connection racing

Team AlgoRhythm

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Problem Description

A measurement problem: when IPv6 and IPv4 both exist, which path actually gets the user connected?

THE PROBLEM

- A hostname can resolve to both AAAA (IPv6) and A (IPv4) addresses, but the two paths may not have equal reachability or latency.
- A client that waits for a slow or broken preferred path can add seconds of avoidable connection delay.
- Happy Eyeballs reduces that wait through staggered connection attempts, yet timing, address ordering, DNS behavior and cancellation can differ across existing clients.
- A successful page load alone does not explain the path taken: the fallback may be invisible without synchronized logs and packet captures.

Research need **Repeatable evidence, not assumptions about undocumented client internals.**

CORE QUESTION

How quickly and reliably do two real clients switch to a usable address family under controlled failure and delay?

WHAT MUST BE MEASURED

- Connection setup time and first response
- Winning family and failed attempts
- Fallback success, attempt timing and loser cancellation

Background

Happy Eyeballs is a connection-establishment strategy, not a guarantee that IPv6 will always win.

01

DNS answers

Query A and AAAA promptly; do not let one slow family block all progress.

02

Address order

Apply destination-address policy and interleave families where appropriate.

03

Start first path

Launch the preferred candidate as soon as usable address data is available.

04

Race alternatives

Start another candidate after a bounded delay if no connection succeeds.

05

Keep the winner

Use the first established connection and cancel outstanding attempts.

WITHOUT EFFECTIVE FALLBACK

IPv6 SYNs stall or fail

long timeout

IPv4 begins only after the wait

WITH STAGGERED RACING

IPv6 starts first **IPv4 starts after delay**



First successful connection wins; pending attempts are cancelled.

RFC 8305 reference value: 250 ms recommended connection-attempt delay (Implementations may differ according to client). Address-selection policy and DNS timing must be recorded.

Literature & RFC Review

Normative baseline first; evolving drafts and diagnostic tooling inform how the experiment is observed.

SOURCE / STATUS	RELEVANT CONTRIBUTION	HOW THIS PROJECT USES IT
RFC 8305 (2017) **Standards Track**	Current normative baseline: asynchronous A/AAAA resolution, destination sorting/interleaving, staggered attempts, and cancellation of losing attempts.	Pin as the primary reference; record actual client timings rather than assuming defaults.
RFC 6724 (2012) **Address selection**	Host policy influences destination-address ordering before connection racing; OS configuration can change the observed result.	Freeze OS and address-selection configuration; preserve resolved answer order.
IETF 126 HAPPY session **July 2026**	The working-group session listed a Happy Eyeballs Webtester presentation, showing current measurement/tooling activity.	Use as related measurement context; the harness still needs pinned conditions and raw evidence.
Happy Eyeballs v3 **draft -04 (1 Jul 2026)**	Active IETF Internet-Draft describing updated concurrent connection behavior; draft status is not the same as a published RFC.	Pin exact revision at project start; do not claim v3 conformance unless status and client support are verified.
Instrumentation **client logs + PCAP + netem**	Chromium NetLog and Firefox networking logs expose diagnostic events; Linux netem can emulate delay/loss; packet capture independently validates traffic.	Correlate evidence from separate layers; do not depend on undocumented internals alone.

Research gap Real-client behavior must be measured across controlled failures with DNS, client logs and PCAPs aligned on one timeline. Passing this experiment is not a standards-conformance certification.

Proposed Solution

A small, isolated dual-stack lab produces one auditable record per client × scenario × repetition.

1 ORCHESTRATOR

Scenario YAML
Randomized run order
Cache/reset control
Monotonic timestamps

2 REAL CLIENTS

Chromium + Firefox pinned
Fresh profile / caches reset
Network logs enabled
QUIC / HTTP/3 disabled

3 CONTROLLED PATHS

Fixed DNS A + AAAA
IPv4 / IPv6 impairment
Delay or blackhole rules
Reversible per-run state

4 DUAL-STACK ORIGIN

Same hostname and content
IPv4 + IPv6 listeners
Fixed TLS certificate / key
Stable response body

EVIDENCE COLLECTOR + NORMALIZER

Chromium NetLog • Firefox network logs • tcpdump/Wireshark PCAP • origin access log

Join with run ID, connection tuple and monotonic timestamps; PCAP validates attempts while client logs explain diagnostic events.

NORMALIZED OUTPUTS

timeline.csv	DNS / SYN / handshake / first byte
attempts.jsonl	Every v4/v6 attempt + failure reason
pcap/ + logs/	Raw evidence retained per run
report + plots	Median latency, fallback and win rates

Experimental Design: Six Controlled Scenarios

Each scenario is run on both pinned clients with the same DNS, origin content and trial protocol.

ID / SCENARIO	CONTROLLED CONDITION	MEASUREMENT PURPOSE
S1 Baseline	IPv4 and IPv6 both healthy; matched low baseline RTT and low jitter.	Establish per-client reference latency, winning-family distribution and extra-attempt rate.
S2 IPv6 blackhole	Drop IPv6 packets to the origin; keep IPv4 healthy.	Measure time to IPv4 fallback, timeout behavior and whether the page still succeeds.
S3 IPv6 slow path	Add 400 ms RTT to IPv6; keep IPv4 on baseline delay.	Test whether racing avoids waiting for a slower initially preferred path.
S4 IPv4 blackhole	Drop IPv4 packets to the origin; keep IPv6 healthy.	Check that a working IPv6 path can complete without unnecessary IPv4 waiting.
S5 IPv4 slow path	Add 400 ms RTT to IPv4; keep IPv6 on baseline delay.	Measure family selection under the reciprocal delay condition.
S6 Delayed AAAA	Delay AAAA DNS response by 500 ms; return A promptly; hold TTL and records fixed.	Observe DNS-resolution behavior when IPv4 data arrives first; compare with RFC 8305 guidance.

RUN PROTOCOL

20 repetitions per client per scenario • randomized order • fresh browser profile / cleared DNS state • fixed OS, resolver, origin and TLS
Impairments must be applied symmetrically where needed, verified before each trial, and fully removed after the run. The specified delay values are lab settings, not claims about typical Internet RTT.

240 MINIMUM RUNS

2 clients × 6 scenarios × 20 repeats

Measurement, Timeline & Analysis

Use a single monotonic clock and corroborate client-reported events with packet-level evidence.

EXAMPLE EVENT TIMELINE (ILLUSTRATIVE, NOT MEASURED DATA)



Primary metric T_connect = navigation start → winning TCP handshake complete

Report separately: DNS duration, TCP handshake duration, TLS completion, time-to-first-byte (TTFB), losing attempts and cancellation timing.

PER-TRIAL FIELDS

Outcome	HTTP success / error / timeout
Path	Winner family + address; failed attempts
Timing	DNS, first SYN per family, TCP/TLS, TTFB
Reproducibility	Client build, scenario config, seed, run ID
Evidence	PCAP + client logs + origin log paths

ANALYSIS

Per client × scenario: median, IQR and success rate; winning IPv4/IPv6 share; fallback delay; extra connections. Compare medians with 95% bootstrap confidence intervals.

RUNNER LOGIC

for client, scenario, repetition: reset state → apply impairment → start PCAP/logs → navigate → classify winner → save timeline → teardown.

Never infer a winner from latency alone: reconcile client logs with SYN/SYN-ACK/ACK sequences and keep raw evidence for disputed runs.

Validation, Real-World Value, Deliverables & Limitations

The output is a reproducible benchmark package, not an invented performance result or a standards certification.

VALIDATION EVIDENCE

- Every run has scenario ID, repetition, pinned client build and monotonic timeline.
- At least 20 valid repetitions per client × scenario; target 240 completed trials.
- PCAP confirms attempted address family, handshake outcome and cancellation / reset evidence.
- Repeat or label invalid runs when DNS, origin, capture or impairment pre-checks fail.

REAL-WORLD APPLICATION

- Campus and enterprise network teams can diagnose partial IPv6 reachability without disabling IPv6 globally.
- Browser / client QA can compare fallback latency after network-stack updates.
- ISPs and site reliability teams can distinguish DNS delay from transport-path failure.
- The same harness can later add loss, jitter, VPN, NAT64 or HTTP/3 as separate experiments.

LIMITATIONS & SAFEGUARDS

- Two pinned clients and one lab topology do not represent every OS, browser or ISP.
- Baseline isolates TCP-based web setup; HTTP/3 / QUIC should be a separately controlled extension.
- Do not rely on undocumented client internals or claim conformance from observed outcomes.
- Network logs may contain sensitive URLs; use a private test hostname and redact before sharing.

IMPLEMENTATION SEQUENCE

1. **Build lab** — DNS + origin + client pins
2. **Calibrate** — Prove each impairment in PCAP
3. **Measure** — Run 240+ trials and validate
4. **Report** — Dataset + plots + demo

Deliver: scenario runner + pinned configuration • two client profiles • six-scenario dataset • packet captures and client logs • analysis notebook / plots • technical report • mentor demo with pseudocode.

References

Primary standards and official tooling documentation checked for this proposal on 13 October 2026.

[1] **AIORI Hackathon 3 - PS-028 problem statement**

Organizer-supplied brief, “Happy Eyeballs IPv4/IPv6 Fallback Measurement,” pp. 1-2.

Source: supplied problem statement PDF

[2] **RFC 8305 - Happy Eyeballs Version 2**

D. Schinazi and T. Pauly, IETF Standards Track, December 2017. Current algorithm baseline.

https://www.rfc-editor.org/rfc/rfc8305.html

[3] **RFC 6724 - Default Address Selection for IPv6**

D. Thaler et al., September 2012. Address ordering and local policy context.

https://www.rfc-editor.org/rfc/rfc6724.html

[4] **IETF 126 HAPPY working-group session**

July 2026 session agenda includes a Happy Eyeballs Webtester presentation; related measurement activity, not a normative standard.

https://datatracker.ietf.org/meeting/126/session/happy

[5] **Happy Eyeballs Version 3 - draft-ietf-happy-happyeyeballs-v3-04**

T. Pauly et al., IETF Internet-Draft, updated 1 July 2026. Re-check publication/status at experiment launch.

https://datatracker.ietf.org/doc/draft-ietf-happy-happyeyeballs-v3/

[6] **Chromium NetLog documentation**

Official Chromium networking event logging reference. Treat logs as diagnostics, not a stable public API.

https://chromium.googlesource.com/chromium/src/+/main/net/docs/net-log.md

[7] **Firefox HTTP / networking logging**

Firefox Source Docs: capture network logs to diagnose connection behavior.

https://firefox-source-docs.mozilla.org/networking/http/logging.html

[8] **Linux tc-netem manual**

Network emulation for delay, jitter, packet loss and related controlled impairments.

https://man7.org/linux/man-pages/man8/tc-netem.8.html

Draft-status rule: freeze the exact v3 revision and verify the IETF Datatracker again on the first day of testing.